Response to Reviewers

Journal: International Journal of Intelligent Engineering and Systems (IJIES) **Paper ID:** 20266428 **Title:** DEAC: A Defensive Ensemble Framework for Automatic Graph-Based Clustering of High-Dimensional Data

We thank both reviewers for their careful and constructive reviews. The common, central point — that our headline comparisons relied on a *label-selected oracle* while the paper's actual product is a *label-free* rule — was well taken, and it drove a substantial revision. We have **re-centered the entire evaluation on the deployable, label-free QualityGate**, regenerated every table and figure from a **single canonical run** (five seeds, full data), and reframed the paper around its genuine contribution: the internal–external misalignment finding and the oracle–deployable gap. We now claim *competitive, count-free* performance rather than superiority. All per-seed results and scripts are released.

Summary of major changes

1. The **deployable gate is now the primary evaluated method** everywhere (main results, comparisons, ablations, significance); the oracle appears only as a clearly labeled diagnostic upper bound.
2. **One canonical source** (5 seeds, full data) produces all numbers; a full-precision per-seed matrix and all scripts are released (GitHub).
3. **New analyses:** gate-constant sensitivity + a development/evaluation leakage test; controlled single-variable ablations; two recent baselines (DPC-AKNN, PRCut); a TOST equivalence test vs. N2D; and a robustness study.
4. **Honest repositioning** of the abstract, contributions, results, and conclusion.

Below, each comment is answered with the change made and where it appears.

Reviewer 1

R1.1 — Figure 6 showed ARI>1, a GROUSE-MH legend, HAR/Letter datasets, and mismatched sample counts. These were stale assets from an earlier experimental iteration, inadvertently retained. All cross-dataset figures have been **regenerated from the final 12-benchmark canonical outputs**; ARI is in [0,1] throughout, only the twelve benchmarks appear, sample counts match Table (Datasets), and the panels show the actual per-dataset ARI, gate activation, and improvement. (Fig. "Main experimental results".)

R1.2 — Inconsistent seeds (42 vs 1001–1005); report the gate and all comparators over the same seeds with dispersion; the 13.1% is the oracle figure. A single protocol is now used: **five fixed seeds (1001–1005), full data, for every method**, with dispersion. The

"seed 42" reference is removed. The abstract is restated around the **deployable** numbers; the oracle is reported separately and explicitly labeled an upper bound. (Secs. "Evaluation Protocol", "Main Results".)

R1.3 — §3.6 gave the ISOLET baseline as C=8 while Table 10 gave 24; Figure 9 counts disagreed. Corrected. The cluster-count table is regenerated from the canonical run, and the rationale now uses a **genuine, canonical-backed under-segmentation case — Olivetti, baseline C≈5 vs. 40 true classes** — instead of the erroneous ISOLET figure. The count table and figures now come from the same run. (Sec. "Design Rationale"; Table "cluster counts".)

R1.4 — Recompute win summaries from unrounded results; state the tie convention; the Friedman rank is inconsistent. We release the **full-precision per-seed ARI matrix** (`canonical_matrix.csv`). All win counts and significance are recomputed from it; a method is credited a win only on a strict row-maximum. The earlier oracle-based Friedman rank is removed; significance is now dataset-level Wilcoxon with Holm–Bonferroni on the deployable gate. (Sec. "Statistical Significance"; Table "stats".)

R1.5 — Figure 7 caption ("9 of 12") contradicted best-on-all-12; some bars (Olivetti baseline ≈0.34 vs 0.076) were wrong; separate wins/ties and oracle from gate. The caption is corrected (no "9 of 12" claim), the figures are regenerated with correct values (Olivetti baseline ≈0.06), and the main results table now **separates the three engines, the deployable gate, and the oracle** in distinct columns. (Fig. "metric bars"; Table "Main Results".)

R1.6 — Figure 10 trajectories reach ~generation 13 though G=8; reconcile with the P·G·R budget and E≈200. The DE configuration is now stated consistently across the text and the figures as **G=8 generations, population P=20 (the LHS-initialised population over the four-dimensional search), R=2 restarts**. The trajectory text and caption state that the horizontal axis **concatenates the R=2 restarts** (hence it exceeds generation 8) before Nelder–Mead refinement, and the evaluation budget is reconciled as $E = R \cdot G \cdot P \approx 320$ ($= 2 \cdot 8 \cdot 20$) in the complexity table, replacing the earlier "E≈200". The AMR-DE schematic (Fig. 3) now shows the same P=20. (Secs. "Evaluation Protocol", "Optimization Dynamics"; Table "complexity"; Fig. 3.)

R1.7 — Table 13 labels t-SNE $O(n^2)$ yet the total sub-quadratic; provide measured timing and peak memory. The complexity table is corrected: t-SNE uses the Barnes–Hut approximation (output dimension ≤3), which is $O(n \log n)$; the sub-quadratic total then holds, with the $O(nk^2)$ SNN term dominating. We add **measured end-to-end wall-clock and peak memory** for the full three-engine pipeline: Wine (n=178) 29 s / 0.6 GB, MFeat (n=2000) 102 s / 0.7 GB, USPS (n=9298) 200 s / 1.8 GB. (Sec. "Computational Complexity"; Table "complexity".)

R1.8 — Specify the executable algorithm; "hybrid candidate H"; r as pruning vs resolution. The parameter-definition conflict is resolved in **both the text and the schematics**: Figures 3 and 4 now label **q as the pruning quantile and r as the Leiden**

resolution (the earlier "r = pruning ratio" / "q = quantile level" labels are removed), matching the released code (`fast_amr_de.py`, where `vec[2]` is the pruning quantile and `vec[3]` is the resolution). The "hybrid candidate H" is now defined explicitly where the gate is introduced: **B is the baseline partition and H is whichever engine output (AMR-DE or EMR-CGC) scores higher internally; the gate accepts H only if it clears all three layers, else falls back to B.** The gate description matches the **implemented deployable form** — among ρ-eligible engines, select the highest internal score (0.40 MOD + 0.35 SIL + 0.25 BAL) — with the Δs and regression checks as conservative safeguards that rarely bind (sensitivity analysis). The exact score normalisations, search bounds, integer rounding, graph bridging, and small-community merging are in the **released reference implementation**. (Secs. "QualityGate", "Design Rationale"; Figs. 3–4; code repository.)

R1.9 — Archive machine-readable artifacts to reconcile outputs. We release, from a single canonical run: a **full-precision per-seed matrix** of internal scores, cluster counts, and gate decisions (`canonical_matrix.csv`); all per-seed result files; and every script that generates each figure, table, and statistical test. The claim in the manuscript is stated to match exactly what is released. (Code & data availability statement; repository.)

Reviewer 2

R2.1 — Make the frozen deployable rule the primary evaluated algorithm; rewrite the abstract and conclusion around it. Done, as the central revision. The deployable gate is the "DEAC" column in the main results, the classical and deep comparisons, the ablations, and the significance tests; the oracle is a labeled diagnostic upper bound only. The abstract and conclusion are rewritten around the deployable results (gate mean ARI 0.623, recovering 65% of the oracle's gain over the baseline; EMR-CGC 0.631). We state plainly where the deployable method underperforms. (Throughout.)

R2.2 — Justify the gate constants; disclose any label influence; dev/eval split; sensitivity; failure cases. We add a sensitivity analysis and a dataset-level development/evaluation split. Mean ARI is **flat across the rho floor (2–6: 0.617–0.624) and c (25–150: 0.624)**, and across the full weight simplex stays in [0.571, 0.627] with the default within 0.003 of the best. Crucially, **tuning the weights on a 6-dataset development split and evaluating on the disjoint six recovers only 40.8% of the oracle's gain vs. 64.9% for the fixed default** (−24.2 points) — fitting the constants to data generalizes *worse*, so they carry no label-dependent advantage. Per-dataset failure cases (the Wine regression; the gate below EMR-CGC in aggregate) are reported. (Sec. "Design Rationale"; Table "sensitivity".)

R2.3 — Isolate the ensemble/SNN/detector contributions with controlled, equal-budget substitutions; a best-internal-score selector and gate-layer removals; the DE engine's deployable value. Two controlled studies are added. (i) A **single-variable**

ablation at matched budget: PCA→UMAP +0.251 mean ARI (representation dominates), Louvain→Leiden −0.003, KNN→SNN −0.010 — so the gain is not attributable to Leiden or SNN per se (SNN helps only on high-dimensional data). (ii) A **selection-rule** ablation: removing the ρ guard leaves the gate unchanged; "always-EMR-CGC" (0.631) in fact exceeds the gate (0.624); dropping AMR-DE from the gate costs only +0.003. We now state that the gate is protective, not accuracy-maximizing, and that AMR-DE does not earn its cost deployably. EMR-CGC is described as selecting the single best-scoring partition (not a consensus). (Secs. "Ablation Study", "Why Arbitration"; Tables "controlled", "selection".)

R2.4 — Add recent methods (DPC-AKNN, PRCut); separate known-C from unknown-C; common-representation vs native pipelines. Both recent methods are added, each **given the true cluster count** (our gate/EMR-CGC infer it). **DPC-AKNN** [23] is faithfully reimplemented from the paper (validated: ARI 1.0 on two-moons/circles), native raw-feature geodesic pipeline → mean ARI 0.420. **PRCut** (AISTATS 2025) is run with the **authors' released implementation** on the **same UMAP representation** as our engines (a controlled common-representation comparison) → mean ARI 0.631. The deployable gate (0.623) and EMR-CGC (0.631) are on par with PRCut (0.631), close to N2D (0.642), and well above DPC-AKNN (0.420) — while inferring C. (Sec. "Comparison with Recent Methods"; Table "recent".)

R2.5 — p=0.622 is not equality; report effect sizes, uncertainty, an equivalence margin; dataset-level vs per-dataset significance; robustness. (i) The "tied with N2D" wording is replaced by a quantified statement: gate − N2D = −0.019 mean ARI (95% CI [−0.079, +0.040]; Wilcoxon p=0.73; Cohen's d_z=−0.18), and a **TOST equivalence test does not establish equivalence even at a ±0.05 margin (p=0.17)** — we claim neither superiority nor equivalence. (ii) All significance statements are now explicitly dataset-level Wilcoxon/Holm; per-dataset win counts are reported as wins, not significance. (iii) A **robustness study** on controlled synthetic data shows the method is robust to class imbalance (ARI 0.49–0.60 up to 30×), improves with separation and with larger n (0.32→0.79 as n:1000→10000), and degrades sharply only under many irrelevant dimensions — stated as an explicit scope limitation. (Secs. "Statistical Significance", "Robustness", "Limitations"; Table "robustness".)

Additional Comments (Editorial)

A1 — Tables and figures must be cited, provided with legends, and numbered serially. Every figure and table is now **cited in the running text** (Fig. 1–11 and Table 1–19 are each referenced where they are discussed), carries a **legend**, and is **numbered serially**. Figure captions appear centred below the figure and table titles centred above the table, in the "Fig. N." / "Table N." form. In-text reference citations, which were missing from the Word export, are restored and numbered ([1]–[34]).

Formatting

The manuscript has been reformatted to the official **IJIES template (Ver. 2026.10.1, `IJIES_Format.docx`)**: A4 two-column layout within the prescribed print area; Times New Roman throughout (14 pt title, 11 pt author, 11 pt body, 10 pt abstract/keywords/captions, 12/11/10 pt headings); numbered first-order headings with a period; the required **Conflicts of Interest** and **Acknowledgments** statements; and the IJIES reference style. No figure, table, or equation extends beyond the print area.

We believe these revisions address every point and make the paper's evidence internally consistent and reproducible. We thank the reviewers again for feedback that materially improved the work.